RCA & RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Fleet Trust Recovery After Cert Rotation

KB 402063 verbatim procedure: clearing a stale VCF Operations integration from Fleet after a certificate rotation breaks the trust handshake.

KB 402063 | FLEET MANAGEMENT | CERT ROTATION | API DELETE | PKIX

VCF 9.0.1 VMware Cloud Foundation

Fleet Trust Recovery After Cert Rotation — KB 402063 Verbatim Procedure

Prepared by: Virtual Control LLC **Date:** May 17, 2026 **Document Version:** 1.0 **Classification:** Internal — Lab Environment **VCF Version:** 9.0.1.0 **Fleet Management:** vmlcm-service-9.0.1.0-SNAPSHOT **VCF Operations:** 9.0.1.0.24960351 **Authoritative KB:** [402063](402063) — public companion of restricted KB 410260

Scope. This document is the standalone runbook for the **cert-rotation-triggered variant** of the "VCF Operations Fleet Management is Not Ready" problem. The broader Fleet-vcops integration RCA (instance-limit deadlock, NPE-flood, DB-direct cleanup) lives in `Fleet-Operations-Integration-Recovery-RCA.md` . Use this document when:

- A Fleet appliance certificate has been rotated in the past few hours
- VCF Ops UI shows `VCF Operations Fleet Management is Not Ready` on Fleet Management → Lifecycle
- `web-vcf.log` on vcf-ops shows `PKIX path building failed`
- Fleet UI Components page shows the `operations` row with **Delete greyed out**

If your symptom set does not match the above (e.g., NPE flood, `vropsMasterNodeIP=null` , DELETE fails after API call), use the broader RCA instead.

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1. Executive Summary

When a Fleet (LCM) appliance certificate is rotated on disk and the appliance is restarted, **VCF Operations' Java truststore is not automatically updated** to trust the new chain. VCF Ops continues to call Fleet over HTTPS with a stale trust state, and every API call from the `ops-lcm` plugin fails with `sun.security.validator.ValidatorException: PKIX path building failed`. The plugin surfaces this to the user as `VCF Operations Fleet Management is Not Ready` on the Lifecycle → VCF Management page — while the Admin → System Status page misleadingly still shows `Connected`.

The supported recovery path is **NOT** to import the new cert into the VCF Ops truststore (no public tooling for this), and **NOT** to revert the Fleet cert to the prior `.bak` files (loses any SANs the rotation introduced). The supported path per **Broadcom KB 402063** is:

1. Disconnect from vcops Admin UI to clear vcops side of the stale 1:1 link
2. Delete the stale environment record on Fleet via API (with `deleteFromVcenter: false`)
3. Reconnect from vcops Admin UI — a fresh registration handshake imports the new cert thumbprint into vcops' trust state

The fix is one API call, gated by a UI-level disconnect on the vcops side and a UI-level reconnect after. The hard part is recognizing that **Fleet UI → Components → Delete is greyed out by design in 9.0.1** and that **KB 410260 is restricted but KB 402063 contains the same procedure publicly**.

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2. Environment Reference

Component	Value (lab)
VCF Operations	`vcf-ops.lab.local` — 192.168.1.77
VCF Operations collector	`collector.lab.local` — 192.168.1.79
Fleet Management (LCM)	`lcm.lab.local` — 192.168.1.95
vCenter	`vcenter.lab.local` — 192.168.1.69
DNS	192.168.1.230
Fleet appliance architecture	Photon 5 hybrid: classic `vrlcm-server.service` (Java, port 8080, `vmlcm-service-9.0.1.0-SNAPSHOT.jar`) PLUS `kind-bootstrap-control-plane` Docker container hosting 27 `vmsp-platform` pods
Fleet truststore (outgoing)	`/var/lib/vrlcm/bundle/fips/cacerts.bcfks` (BCFKS, pw `changeit`)
vcops truststore (Java)	Maintained by vcops cluster; no public re-import procedure

Substitute your own values throughout.

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3. Symptom Signature

Surface	Observation
vcops UI → Fleet Management → Lifecycle → VCF Management	`VCF Operations Fleet Management is Not Ready`
vcops UI → Admin → System Status	`Fleet Management: Connected` (green) — misleading; the admin-level link record still exists
`/storage/log/vcops/log/web-vcf.log`	`Caused by: sun.security.validator.ValidatorException: PKIX path building failed: sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification path to requested target`
`/storage/log/vcops/log/product-ui/localhost_access_log.*.txt`	Repeating `GET /vcf-operations/plug/ops-lcm/lcm/bootstrap/api/status HTTP/2.0 400` every 5s
Fleet UI (break-glass) → Components	`operations` row, status `New Deployment`, **Delete option greyed out**
Fleet API — `GET /lcm/lcops/api/v2/environments`	One env present, `environmentStatus: COMPLETED`, env ID is a UUID
Fleet log `/var/log/vrlcm/vmware_vrlcm.log`	No active errors; cert rotation appears as `RequestProcessor` entries for `Replace Certificate` flow

> If `web-vcf.log` shows a different exception (e.g., `Connection reset`, `UnknownHostException`, `Connection refused`), this is not the cert-rotation case. See sibling RCAs.

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4. Root Cause

Fleet's appliance certificate is presented by `nginx` on port 443 and is loaded from `/opt/vmware/vlcm/cert/server.crt`. When this file is replaced and Fleet is restarted, the new chain is served immediately. VCF Operations, however, holds the **prior chain's public key** in its Java truststore from the original handshake at registration time. Java's PKIX path builder requires the issuer (or self-signed CN) to be in the truststore; the new cert is not there, so every TLS call from the `ops-lcm` plugin fails with `unable to find valid certification path to requested target`.

The plugin retries every 5 seconds and surfaces the failure as `VCF Operations Fleet Management is Not Ready`. Meanwhile, the Admin → System Status page reads the registration **record** (not the live trust state), so it shows `Connected`. The two views disagree because they query different state.

Fleet enforces a strict 1:1 limit: only one VCF Operations instance may be registered per Fleet appliance. After cert rotation, the prior record is **still in Fleet's `vrlcm` PostgreSQL DB** and consumes the only allowed slot. Any reconnect attempt from vcops fails because Fleet rejects the new registration ("Deployment or Import Blocked: You've reached the limit of allowed instances for VCF Operations (maximum: 1)").

The fix is to **delete the stale Fleet-side env record** so the slot frees, then reconnect.

Root Cause Statement. Cert rotation on Fleet invalidates the cert chain held in vcops' Java truststore. The 1:1 registration limit on Fleet blocks a fresh registration until the stale record is cleared. KB 402063 documents the cleanup via API DELETE with `deleteFromVcenter: false` — the load-bearing flag that preserves the actual vcops and collector VMs.

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5. What Does Not Work (and Why)

Action	Result	Why
Reboot Fleet appliance (KB 429291)	Pods return to 27/27 Running; symptom unchanged	KB 429291 fixes **sync-state** mismatches, not **trust-state** mismatches
`REPLACE` button in vcops Admin → System Status	`Error connection Fleet management node, try again` (KB 404388)	1:1 limit blocks overlapping registration; failed attempt leaves a partial entry compounding the problem
Fleet UI → Components → 3-dot menu → Delete	Greyed out, no tooltip	9.0.1 architecture refuses to delete the only/last integrated component via UI
Fleet UI → Components → Manage → "..." top toolbar	Menu items: Vertical Scale Up / Update NTP / Add New Collector Group / Patches / Export Configuration / Snapshot / Logs — **no Delete**	Same 9.0.1 UI constraint
Restart `vrlcm-server.service` on Fleet	No effect on trust state	The trust state is held on the vcops side; Fleet-side restart does not refresh it
Revert Fleet cert from `.bak` and reboot Fleet	May work IF vcops side still trusts the old chain	Loses any SANs/expiry the rotation introduced; usually not desirable
Import new cert into vcops Java truststore directly (`keytool`)	No public tooling/path; modifies state vcops doesn't expect	Not Broadcom-supported
Open Broadcom Support case with KB 410260	Eventually works	Slow (case time); unnecessary when KB 402063 procedure is sufficient

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6. Recovery Procedure (Verbatim)

6.1 Phase 1 — Pre-Recovery Snapshots

Snapshot **all three** appliances before any destructive action. PowerCLI is the fastest path; vCenter UI works equally well.

```
# Run from Windows workstation
Set-PowerCLIConfiguration -InvalidCertificateAction Ignore -Confirm:$false -Scope Session | Out-Null
$cred = New-Object System.Management.Automation.PSCredential('administrator@vsphere.local',
        (ConvertTo-SecureString '<your-password>' -AsPlainText -Force))
Connect-VIServer -Server 192.168.1.69 -Credential $cred -Force

$ts = (Get-Date -Format 'yyyy-MM-dd')
foreach ($name in 'fleet','vcf-ops','collector') {
```

```
  New-Snapshot -VM $name `
               -Name "pre-KB402063-$ts" `
               -Description "KB 402063 Fleet trust recovery prereq snapshot" `
               -Memory:$false -Quiesce:$false -Confirm:$false
}

Get-VM -Name 'fleet','vcf-ops','collector' | Get-Snapshot | Format-Table VM, Name, Created, SizeGB,
IsCurrent
```

Use `-Quiesce:$false`. VMware Tools' guest-side quiesce hook deadlocks on Fleet because the `kind-bootstrap-control-plane` container holds many open files. A quiesced snapshot will stick at 99% for 10-15 minutes then abort with `connection closed by server` and leave nothing on disk. Non-quiesced (crash-consistent) snapshots are correct for rollback purposes — Photon and the inner K8s cluster crash-recover cleanly on revert.

6.2 Phase 2 — Disconnect via vcops Admin UI

1. Open `https://<vcf-ops-fqdn>/admin/` in a browser
2. Sign in as `admin` (vcops admin, NOT `admin@local`)
3. Left sidebar → **System Status**
4. In the **Fleet Management** card, click **DISCONNECT**
5. Confirm the dialog

After confirmation, the `Fleet Management: Connected` indicator should change to `Disconnected` (or to an empty `Connect` / `Add` button state). The `Fleet Management is Not Ready` banner on the Lifecycle page will remain — that is expected; it will clear after reconnection.

6.3 Phase 3 — Enable Fleet Break-Glass UI

```
 # Run on the appliance only — Fleet (192.168.1.95) as root
touch /var/lib/vrlcm/UI_ENABLED
ls -la /var/lib/vrlcm/UI_ENABLED
# Expected: -rw-r----- 1 root root 0 <today's date> /var/lib/vrlcm/UI_ENABLED
```

Then open `https://<fleet-fqdn>` in a browser and sign in as `admin@local` with Fleet's admin@local password.

Why the break-glass UI? While the API DELETE in §6.4 can be issued from any host with `curl`, sourcing the **environment ID** is most reliably done from the Fleet UI's Components page or the Fleet API's `/lcm/lcops/api/v2/environments` endpoint. The UI also lets you confirm the post-delete state visually (the row disappears).

6.4 Phase 4 — KB 402063 API DELETE

6.4.1 Find the environment ID

```
 # Run on Fleet (.95) or any host with curl + admin@local credential
curl -sk -u admin@local:'<password>' https://localhost/lcm/lcops/api/v2/environments \
  | python3 -m json.tool | head -20
# Capture the "environmentId" value of the stale vrops entry, e.g.:
#   "environmentId": "888501ce-eb95-4ce8-8043-ab216906bc67"
```

```
# PowerShell sibling
[System.Net.ServicePointManager]::ServerCertificateValidationCallback = {$true}
[System.Net.ServicePointManager]::SecurityProtocol = [System.Net.SecurityProtocolType]::Tls12
$u='admin@local'; $p='<password>'
$h=@{Authorization=('Basic '+[Convert]::ToBase64String([Text.Encoding]::ASCII.GetBytes("$u`:$p")));
     'Content-Type'='application/json'}
$envs = Invoke-RestMethod -Uri 'https://lcm.lab.local/lcm/lcops/api/v2/environments' -Headers $h
$envs | Select-Object environmentId, environmentStatus
```

6.4.2 DELETE the environment (KB 402063 verbatim payload)

```
ENV="<paste env ID from §6.4.1>"

curl -sk -u admin@local:'<password>' \
  -X DELETE \
  -H "Content-Type: application/json" \
  -d '{
    "controllerType": "string",
    "deleteFromInventory": true,
    "deleteFromVcenter": false,
    "deleteLbFromSddc": true,
    "deleteWindowsVMs": true
  }' \
  "https://localhost/lcm/lcops/api/environments/$ENV/delete"
# Expected response: {"requestId":"<uuid>"}
```

```
# PowerShell sibling
$body = @{
  controllerType      = "string"
  deleteFromInventory = $true
  deleteFromVcenter   = $false
  deleteLbFromSddc    = $true
  deleteWindowsVMs    = $true
} | ConvertTo-Json
$envId = "<paste env ID>"
Invoke-RestMethod -Uri "https://lcm.lab.local/lcm/lcops/api/environments/$envId/delete" `
                  -Method DELETE -Headers $h -Body $body
```

`deleteFromVcenter: false` is load-bearing. With this flag, only the LCM DB record is cleared — the actual `vcf-ops` and `collector` VMs in vCenter are not touched. With `true`, Fleet attempts to power off and delete those VMs. **Confirm the flag value before pressing Execute.**

The `"controllerType": "string"` field is a Swagger placeholder retained verbatim from KB 402063; Fleet's `RequestProcessor` ignores it. Substituting `"VROPS"` or other product strings is not required.

6.5 Phase 5 — Verification

6.5.1 Tail the LCM log for completion

```
# Run on Fleet — tail the LCM log for the requestId returned in §6.4.2
tail -F /var/log/vrlcm/vmware_vrlcm.log | grep -E "<requestId>|environmentdelete"
```

Success log line pattern:

```
<timestamp> INFO vrlcm[<pid>] [scheduling-1] [c.v.v.l.r.c.RequestProcessor]
  -- Updating the Environment request status to COMPLETED for request ID :
  <requestId> with request type : DELETE_ENVIRONMENT.
```

API quirk. `GET /lcm/api/v2/requests/{requestId}` returns `HTTP 500 Internal Server Error` on this build (`{"timestamp":...,"status":500,"error":"Internal Server Error","path":"/lcm/lcops/api/requests/<id>"}`). The DELETE itself succeeds; only the request-lookup endpoint is broken. Use the LCM log for status, not the API.

6.5.2 Confirm env is gone

```
 # Returns []
curl -sk -u admin@local:'<password>' "https://localhost/lcm/lcops/api/v2/environments"

# Returns HTTP 404
curl -sk -u admin@local:'<password>' -o /dev/null -w "%{http_code}\n" \
  "https://localhost/lcm/lcops/api/v2/environments/$ENV"
```

6.5.3 Confirm VMs are still alive

```
 foreach ($name in 'vcf-ops','collector','fleet') {
  $vm = Get-VM -Name $name
  "{0,-12}  Power={1}  Heartbeat={2}  IPs={3}" -f $vm.Name, $vm.PowerState, $vm.Guest.State,
    (($vm.Guest.IPAddress | Where-Object {$_ -match '^\d+\.\d+\.\d+\.\d+$'}) -join ',')
}
# All three should report: Power=PoweredOn  Heartbeat=Running
```

6.6 Phase 6 — Reconnect via vcops Admin UI

1. In vcops Admin UI → **System Status**, click the **Connect** / **Add Fleet Management Node** button (its label varies by minor version)
2. Fill in:

Field	Value
Fleet FQDN	`lcm.lab.local`
Username	`admin@local`
Password	Fleet's `admin@local` password
3. When prompted with the certificate-accept dialog, **verify the SHA-1 thumbprint matches what Fleet is presenting live**:	

`powershell # Pull Fleet's live cert thumbprint $tcp = [System.Net.Sockets.TcpClient]::new('192.168.1.95',443) $ssl = [System.Net.Security.SslStream]::new($tcp.GetStream(),$false,({$true})) $ssl.AuthenticateAsClient('192.168.1.95') $c = [System.Security.Cryptography.X509Certificates.X509Certificate2]::new($ssl.RemoteCertificate) $c.Thumbprint $ssl.Close(); $tcp.Close()`

Compare digit-for-digit with the value shown in the UI accept dialog. 4. Accept the cert and click **Connect** / **Submit**.

Within ~30 seconds the System Status should show `Fleet Management: Connected`; refresh Fleet Management → Lifecycle → VCF Management and the `Not Ready` banner should be gone.

6.7 Phase 7 — Cleanup

```
# Disable break-glass UI
rm /var/lib/vrlcm/UI_ENABLED
```

Plan snapshot consolidation: the chain from §6.1 plus any prior session snapshots can grow large quickly. Consolidate via vCenter UI (right-click VM → Snapshots → Manage Snapshots → Delete All) once Fleet operations have stabilized for 24-48 hours.

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7. Rollback Procedure

If any of Phases 4-6 fails irrecoverably, roll back to the snapshots taken in §6.1.

```
# Revert all three VMs to their pre-KB402063 snapshots
foreach ($name in 'fleet','vcf-ops','collector') {
  $vm = Get-VM -Name $name
  $snap = Get-Snapshot -VM $vm -Name "pre-KB402063-*"
  if ($snap) {
    Set-VM -VM $vm -Snapshot $snap -Confirm:$false
  }
}
```

After revert, the prior state returns: Fleet still has the stale env record, vcops still has the broken trust state. From there, either: - Open a Broadcom Support case citing **KB 410260** for the restricted-access procedure - Retry this runbook with corrected inputs

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8. KB Citation Map

KB	Role in this recovery	Access
402063	**Authoritative DELETE endpoint and payload**	**Public**
410260	Same procedure, support-restricted	Broadcom-Support-only
430561	Procedure outline (disconnect → enable UI → delete → reconnect)	Public; predates 9.0.1 UI greyout
432320	Root-cause framing: 1:1 limit + cert change	Public
429291	Sync-state out-of-sync — insufficient for cert-rotation case	Public
404388	Explains `REPLACE` button error and partial-entry behavior	Public
433975	Cert SAN missing FQDN — ruled out if SANs are correct	Public

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9. Incident Log — 2026-05-17 (First Documented Use)

Property	Value
Trigger	Fleet cert rotation on 2026-05-16 16:48 UTC; new SHA-1 thumbprint `49:B3:F9:95:AA:26:8F:F1:E2:EE:CC:B4:3D:2B:C3:10:0D:D2:D5:E0`

Property	Value
Stale env ID	`888501ce-eb95-4ce8-8043-ab216906bc67`
KB 402063 DELETE requestId	`1772209a-f300-481d-86ae-ce489090f9b7`
DELETE completion timestamp	`2026-05-17T17:12:46.450Z`
Phase 1 (snapshots) duration	~50 min — first attempt with `-Quiesce:$true` stuck at 99% and aborted after 13 min; second attempt with `-Quiesce:$false` completed in ~30 s
Phase 4 (API DELETE) duration	~5 min from issue to completion
Phase 6 (reconnect) duration	~2 min
Net hands-on time	**~15 min once correct KB identified**
Total elapsed (incl. wrong paths tried)	~2.5 hours
Result	Re-connected cleanly; VCFA deploy unblocked

Wrong paths attempted before reaching KB 402063 (all reasonable initial guesses, all insufficient):

Attempted	Outcome
KB 429291 reboot of Fleet	Pods 27/27 OK; symptom unchanged
`REPLACE` button in vcops Admin UI	`Error connection Fleet management node, try again` (KB 404388)
Fleet UI → Components → 3-dots → Delete	Greyed out
Fleet UI → Manage → toolbar `...`	No Delete in menu (only Scale Up / NTP / Patches / Snapshot / Logs)

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10. Lessons Learned

1. The vcops Admin UI status is not a source of truth for plugin trust health. It can show `Connected` while the `ops-lcm` plugin is throwing PKIX exceptions at every request. Always corroborate with `/storage/log/vcops/log/web-vcf.log`.

2. KB 402063 is the public companion of restricted KB 410260. When a recovery procedure is described in a support-only KB, search for its public sibling before opening a case.

3. VCF 9.0.1 Fleet UI's `Components → Delete` is greyed out by design for the only/last integrated component. Do not look for a UI-side force-delete in this build — go to the API.

4. `deleteFromVcenter: false` is the only difference between a safe cleanup and a destructive teardown. This single boolean preserves the actual VMs. Confirm its value in the Swagger "Try it out" body before clicking Execute — Swagger defaults all booleans to `true`.

5. Quiesced snapshots deadlock on K8s/Docker-heavy guests. Use `-Quiesce:$false` for recovery rollback snapshots on Fleet, vcf-ops, and any appliance running an inner K8s cluster.

6. The Fleet API `/lcm/api/v2/requests/{requestId}` endpoint is broken on 9.0.1. Returns HTTP 500 even when the underlying request succeeded. Use `/var/log/vrlcm/vmware_vrlcm.log` to confirm DELETE completion.

7. Multiple KBs share the same error message. "Error connection Fleet Management node, try again" appears in KB 404388, KB 410260, KB 432320 with different root causes. Use Fleet's logs and the cert-rotation timeline to disambiguate.

8. The Admin UI's `REPLACE` button is not a recovery action — it is a node-swap workflow. When stale state exists, REPLACE compounds the problem by leaving a second partial entry (KB 404388). Always Disconnect first.

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11. Document Information

Field	Value
Title	Fleet Trust Recovery After Cert Rotation — KB 402063 Verbatim Procedure
Document ID	VC-RCA-FLEET-CERT-402063
Version	1.0
Issued	May 17, 2026
Author	Virtual Control LLC
Classification	Internal — Lab Environment
Related Documents	`Fleet-Operations-Integration-Recovery-RCA.md` (broader integration RCA; §15 references this document by the same incident date), `VCF-Automation-Deployment-Troubleshooting-Report.md` (VCFA deploy unblocked once this recovery completes), `NSX-Manager-Cold-Start-Troubleshooting-Report.md` (sibling cold-start RCA)
Distribution	Lab notebook; do not redistribute externally without redaction of credentials and hostnames

Revision History

Version	Date	Notes
1.0	May 17, 2026	Initial release. Standalone runbook for cert-rotation-triggered Fleet/vcops trust recovery via KB 402063 API DELETE. Includes verbatim KB payload, PowerShell siblings for workstation-runnable steps, snapshot rollback procedure, and incident log from first documented use (env `888501ce-eb95-4ce8-8043-ab216906bc67`, DELETE requestId `1772209a-f300-481d-86ae-ce489090f9b7`, completed 2026-05-17T17:12:46.450Z).

Document End

See also: - `Fleet-Operations-Integration-Recovery-RCA.md` — the broader Fleet/vcops integration RCA covering instance-limit deadlock, NPE-flood, and DB-direct cleanup paths - `VCF-Automation-Deployment-Troubleshooting-Report.md` — VCFA deployment troubleshooting (the deploy this recovery unblocks) - `NSX-Manager-Cold-Start-Troublesho `oting-Report.md` — sibling cold-start RCA - KB 402063 (public): `https://knowledge.broadcom.com/external/article/402063`